Group 4: Acoustic Filter Design for Silent Temperature Humidity Chamber

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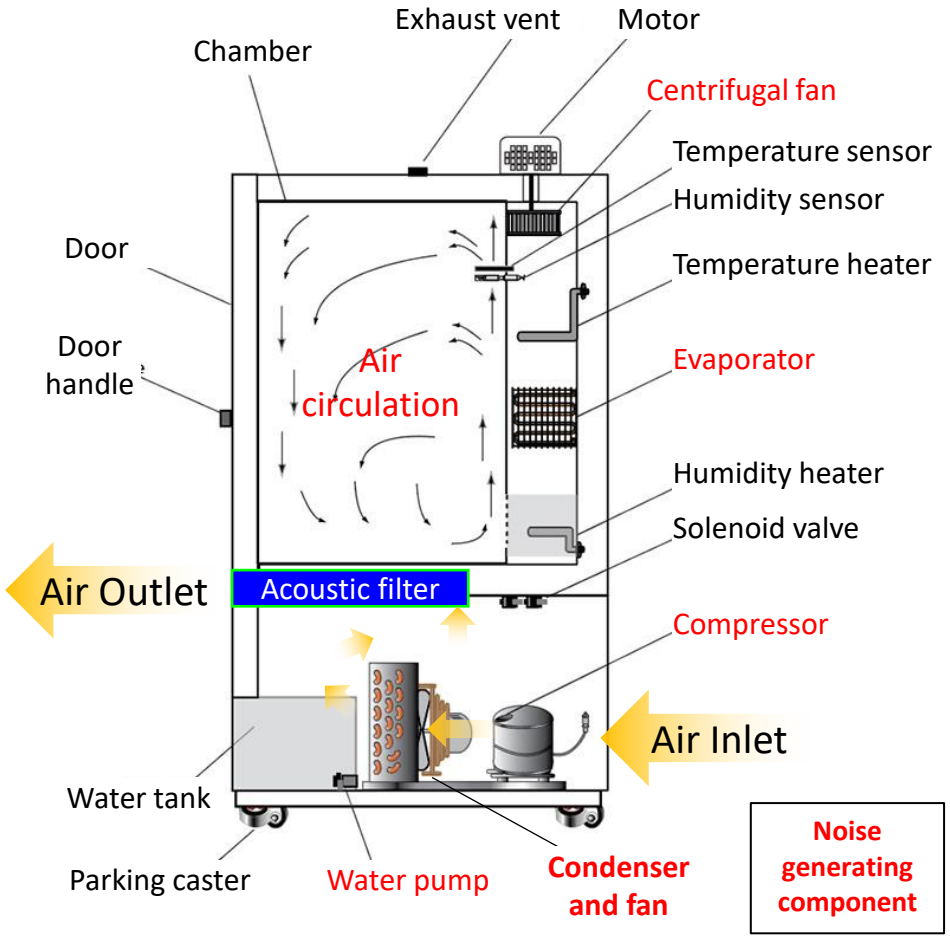
Introduction

- Temperature humidity chamber (climate testing)
- Scenario
  - Main frequency of the condenser fan is around 300 Hz.
  - Other components generate noise in wide frequency range.
- Goal: Design acoustic filters to protect graduate students

Temperature humidity chamber and graduate student



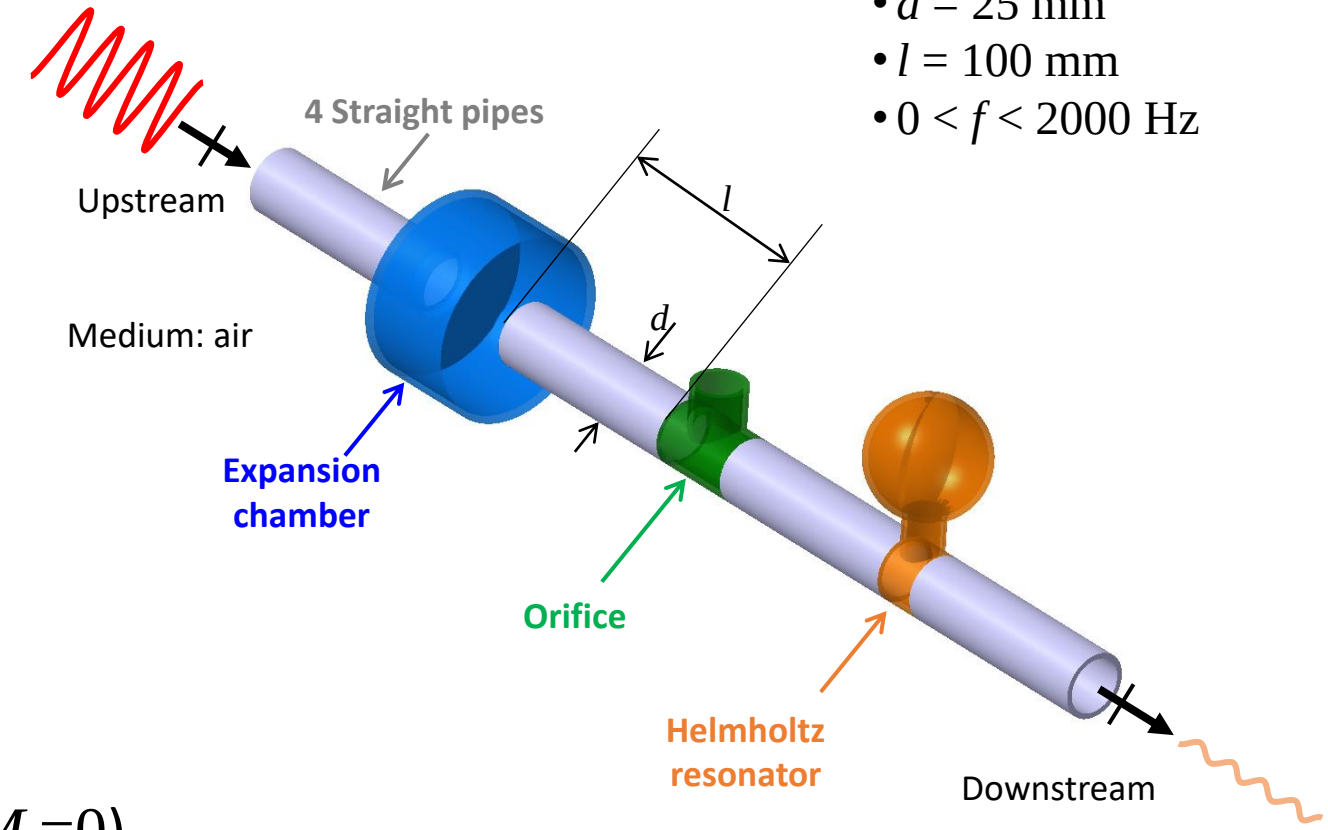
Schematic of temperature humidity chamber

Acoustic filter system

- Design objective
  - Design each acoustic filter element to reduce sound transmission in all given frequency range (0 ~ 2000 Hz).
  - Find the proper order of elements
  - Main target frequency: 300 Hz
- Assumption
  - Medium is air.
  - There is no dissipation.
  - Outlet is anechoically terminated.
  - Plane wave is propagating through the system.
  - Mean flow velocity of air is near zero ($M = 0$).

Given:

- $d = 25$ mm
- $l = 100$ mm
- $0 < f < 2000$ Hz

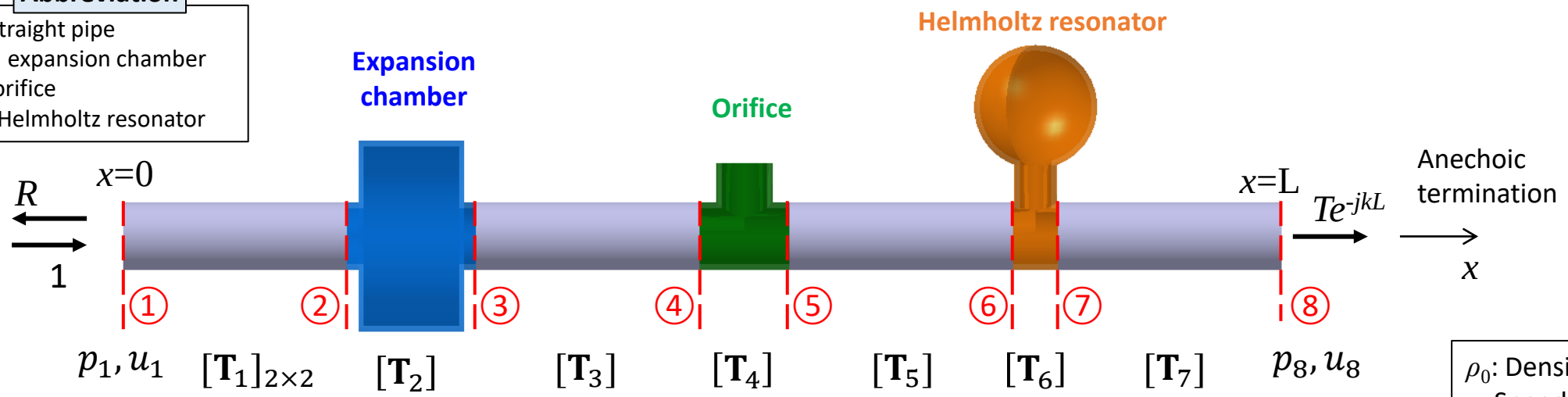


Acoustic filter system

Design of acoustic filter: Transfer matrix approach

Abbreviation

s: straight pipe
e.c: expansion chamber
o: orifice
H: Helmholtz resonator



$$[\mathbf{T}_{1,3,5,7}] = [\mathbf{T}_{\mathrm{s.p}}]$$
$$[\mathbf{T}_2] = [\mathbf{T}_{\mathrm{e}}]$$
$$[\mathbf{T}_4] = [\mathbf{T}_{\mathrm{o}}]$$
$$[\mathbf{T}_6] = [\mathbf{T}_{\mathrm{H}}]$$

$$\begin{bmatrix} p_1 \\ U_1 \end{bmatrix} = \mathbf{T}_1\mathbf{T}_2\mathbf{T}_3\mathbf{T}_4\mathbf{T}_5\mathbf{T}_6\mathbf{T}_7 \begin{bmatrix} p_8 \\ U_8 \end{bmatrix}$$
$$= \mathbf{T}_{\mathrm{s.p}}\mathbf{T}_{\mathrm{e}}\mathbf{T}_{\mathrm{s.p}}\mathbf{T}_{o}\mathbf{T}_{\mathrm{s.p}}\mathbf{T}_{\mathrm{H}}\mathbf{T}_{\mathrm{s.p}} \begin{bmatrix} p_8 \\ U_8 \end{bmatrix}$$
$$= \mathbf{T}_{\mathrm{system}} \begin{bmatrix} p_8 \\ U_8 \end{bmatrix}$$
$$= \begin{bmatrix} T_{11} & T_{12} \\ T_{13} & T_{14} \end{bmatrix} \begin{bmatrix} p_8 \\ U_8 \end{bmatrix} \quad \dots (1)$$

$$\begin{bmatrix} 1+R \\ \frac{S}{\rho_0 c}(1-R) \end{bmatrix} = \begin{bmatrix} T_{11} & T_{12} \\ T_{13} & T_{14} \end{bmatrix} \begin{bmatrix} Te^{-jkL} \\ \frac{S}{\rho_0 c}Te^{-jkL} \end{bmatrix} \quad \dots (2)$$

$$R = \frac{T_{11} + \frac{S}{\rho_0 c}T_{12} - \frac{\rho_0 c}{S}T_{21} - T_{22}}{T_{11} + \frac{S}{\rho_0 c}T_{12} + \frac{\rho_0 c}{S}T_{21} - T_{22}}$$

$$T = \frac{2e^{+jkL}}{T_{11} + \frac{S}{\rho_0 c}T_{12} + \frac{\rho_0 c}{S}T_{21} + T_{22}}$$

$$R_\Pi = |R|^2 \qquad T_\Pi = |T|^2$$

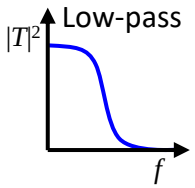
$$R_\Pi + T_\Pi = 1 \qquad TL = 10\log_{10}\frac{1}{T_\Pi}$$

Nomenclature

ρ_0: Density of air
c: Speed of sound
k: Wavenumber
$[\mathbf{T}]$: Transfer matrix
T_{ij}: Element of transfer matrix
p: Pressure
u: Particle velocity
S: Cross-section area
U: Volume velocity (= Su)
R: Reflection coefficient
R_Π: Power reflection coefficient
T: Transmission coefficient
T_Π : Power transmission coefficient
TL: Transmission loss

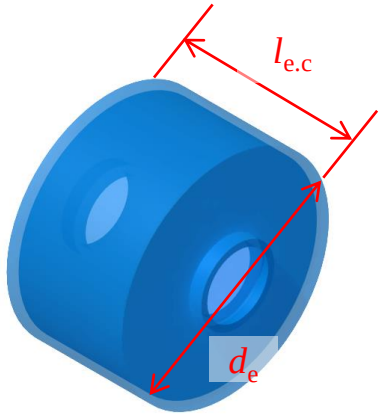
Modeling of transfer matrix of each element

Expansion chamber

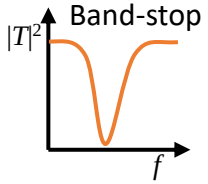


$$[\mathbf{T}_\mathrm{e}] = \begin{bmatrix} \cos(kl_\mathrm{e}) & j\dfrac{\rho_0 c}{S_\mathrm{e}}\sin(kl_\mathrm{e}) \\ j\dfrac{S_\mathrm{e.c}}{\rho_0 c}\sin(kl_\mathrm{e}) & \cos(kl_\mathrm{e}) \end{bmatrix}$$

where $S_\mathrm{e} = \frac{\pi}{4}d_\mathrm{e}^2$



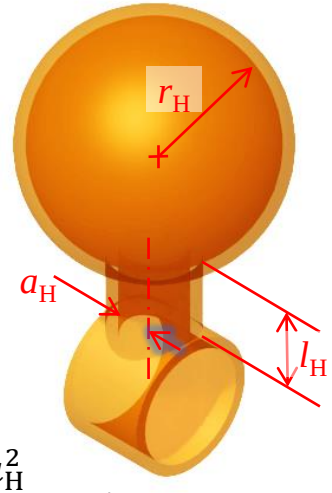
Helmholtz resonator



$$[\mathbf{T}_\mathrm{H}] = \begin{bmatrix} 1 & 0 \\ \dfrac{\pi\omega V a_\mathrm{H}^2}{j\{\rho_0\omega^2 V(l_\mathrm{H} + 1.7a) - \pi\rho_0 c^2 a_\mathrm{H}^2\}} & 1 \end{bmatrix}$$

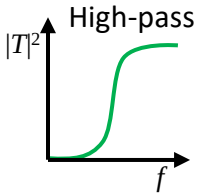
$$Z_b = \cancel{R_m}^{\,0} + j\left(\frac{\rho_0\omega L_\mathrm{H}'}{\pi a_\mathrm{H}^2} - \frac{\rho_0 c}{\omega V}\right)$$

where $L_\mathrm{H}' = l_\mathrm{H} + 1.7a_\mathrm{H}^2$
and $V = \frac{4}{3}\pi r_H^3$



*Flanged end-correction

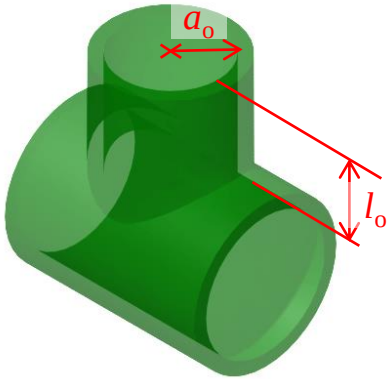
Orifice



$$[\mathbf{T}_\mathrm{o}] = \begin{bmatrix} 1 & 0 \\ \frac{1}{Z_b} & 1 \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ \dfrac{\pi a_o^2}{j\rho_0\omega(l_\mathrm{o} + 1.45a_\mathrm{o}^2)} & 1 \end{bmatrix}$$

$$Z_b = \cancel{\frac{\rho_0 c k^2}{4\pi}}^{\,0} + j\frac{\rho_0\omega L_\mathrm{o}'}{\pi a_o^2}$$

where $L_o' = l_\mathrm{o} + 1.45a_\mathrm{o}^2$

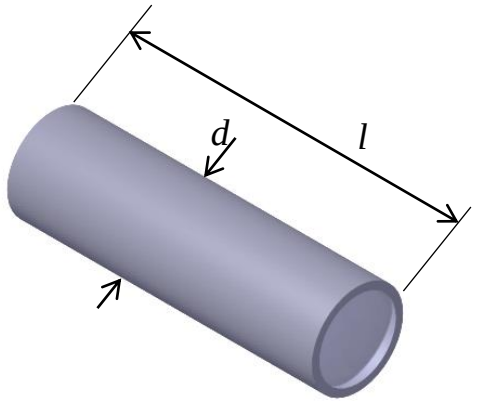


*Unflanged end-correction

Straight pipe

$$[\mathbf{T}_\mathrm{s.p}] = \begin{bmatrix} \cos(kl) & j\dfrac{\rho_0 c}{S}\sin(kl) \\ j\dfrac{S}{\rho_0 c}\sin(kl) & \cos(kl) \end{bmatrix}$$

where $S = \frac{\pi}{4}d^2$

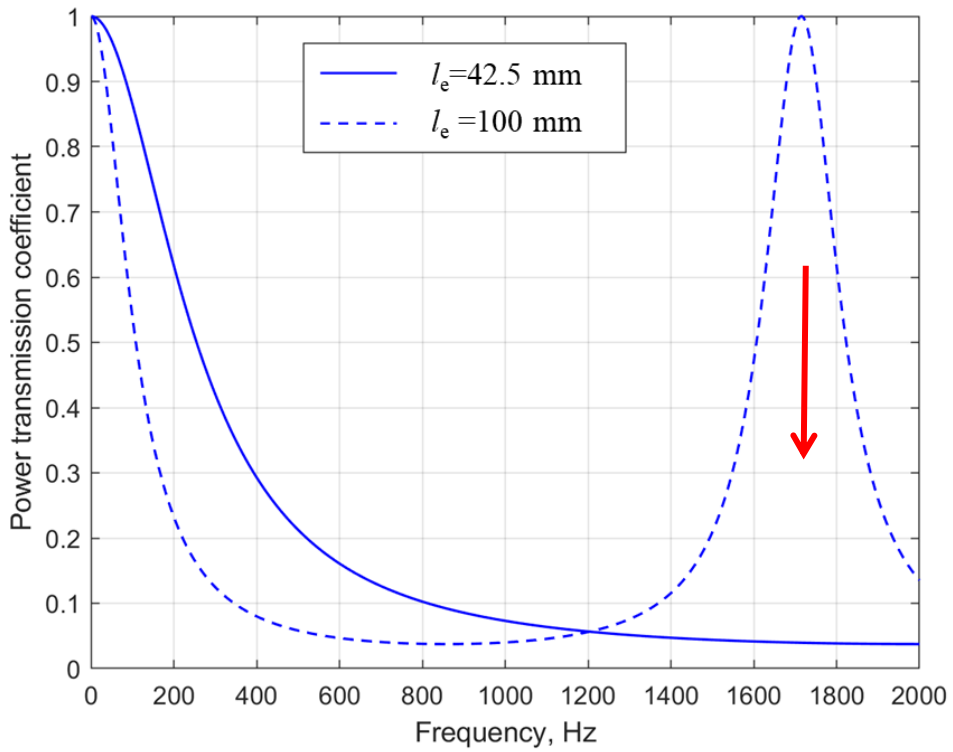


Assumption

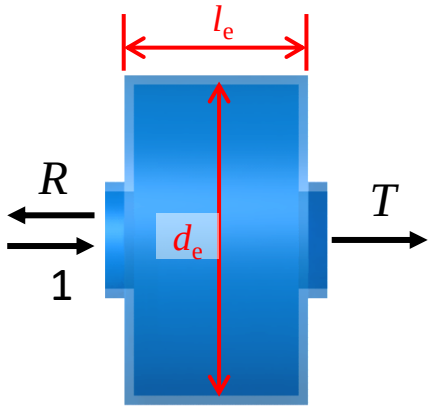
- No mean air flow through expansion and contraction (All elements)
- No viscous & thermal losses and no radiation to infinity (Helmholtz)

Parameter tunning (1): Expansion chamber

Sub-objective: a low-pass filter with wide stop band and high attenuation rate



$$|T| = \frac{2}{\sqrt{4\cos^2(kl_e) + \left(\frac{S}{S_e} + \frac{S_e}{S}\right)^2 \sin^2(kl_e)}}$$



Expansion chamber and parameters

To avoid oscillation:

Let $kl_e \leq \frac{\pi}{2}$ in the frequency range of $0 \sim 2$ kHz

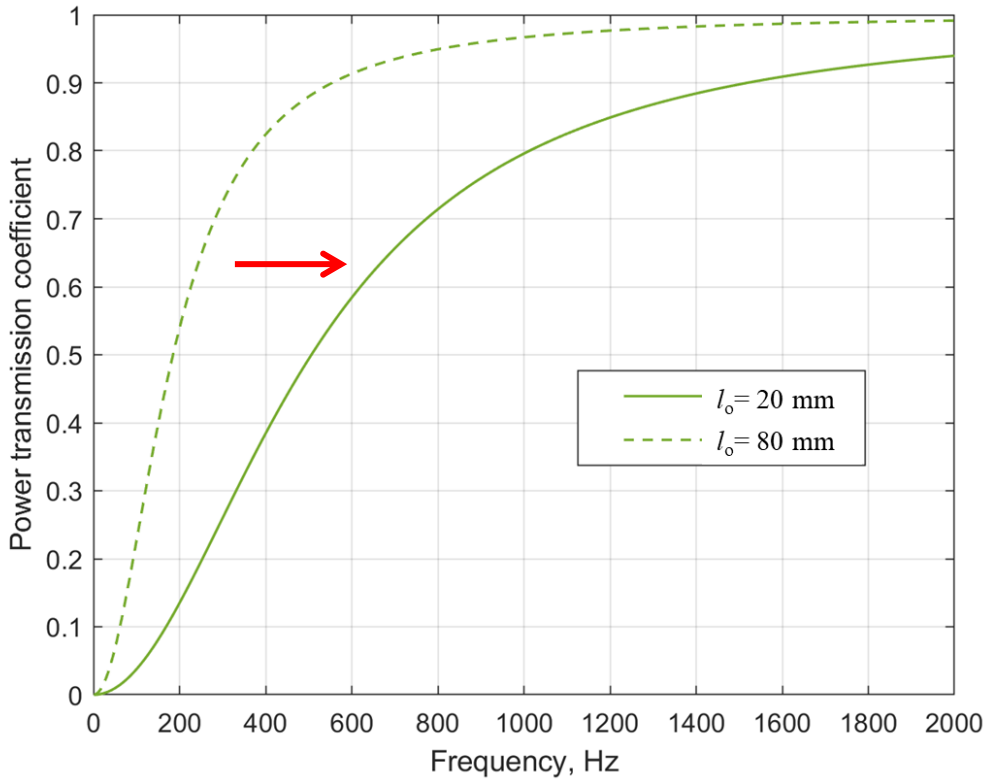
→$l_e \leq 42.5$ mm

For higher attenuation, we also need larger diameter:

→$l_e = 42.5$ mm, $d_e = 80$ mm

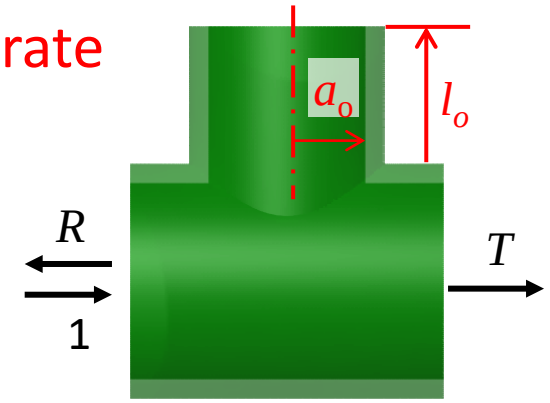
Parameter tunning (2): Orifice

Sub-objective: a high-pass filter with wide stop band and high attenuation rate



$$|T| = \frac{2}{\sqrt{4 + \left(\frac{\pi a_o^2 c}{S(l_o + 1.45a_o)\omega}\right)^2}}$$

$$f_c = \frac{a_o^2 c}{4S\,(l_o + 1.45a_o)}$$



Orifice and parameters

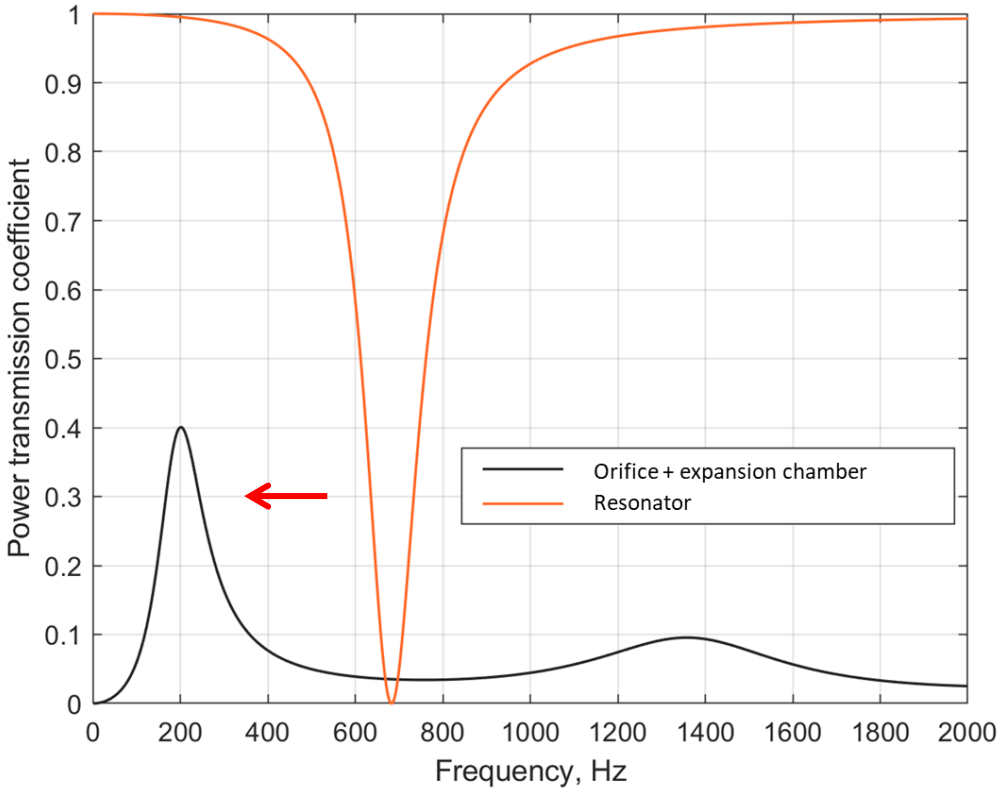
For a wider stop band, we need larger diameter and smaller length:

$a_o = 10\text{ mm}, l_o = 20\text{ mm}$

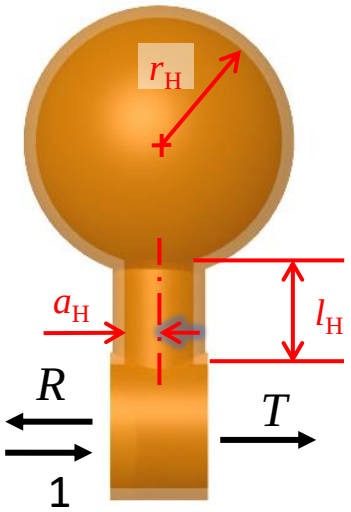
$\rightarrow f_c \cong 500\text{ Hz}$

Parameter tunning (3): Helmholtz resonator

Sub-objective: a band-stop filter to attenuate the bump caused by transition regions of high and low pass filters



$$f_{\mathrm{n}} = \frac{c}{2\pi}\sqrt{\frac{a_{\mathrm{H}}^2}{\frac{4}{3}r_{\mathrm{H}}^3(l_{\mathrm{H}}+1.7a_{\mathrm{H}})}}$$

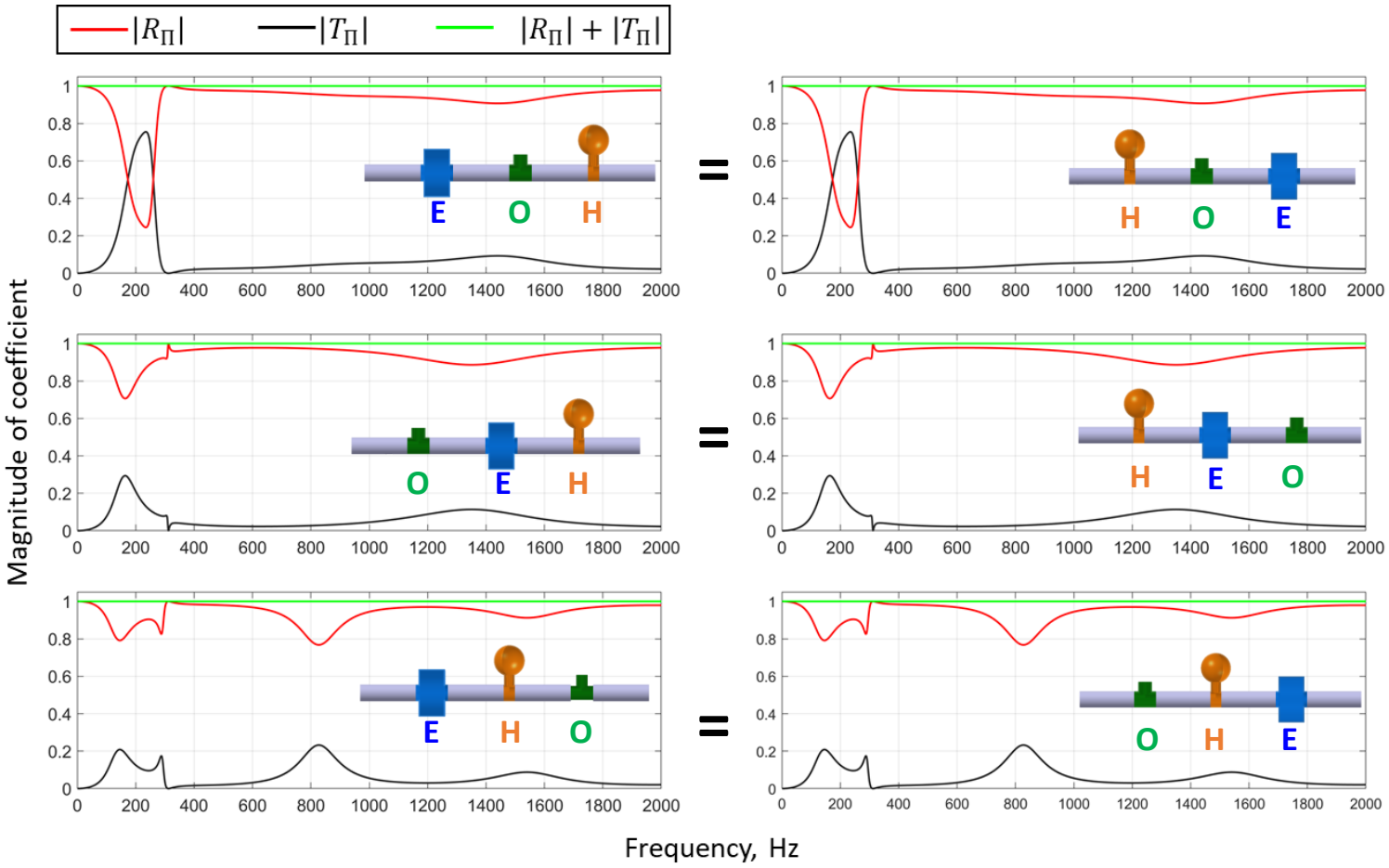


Helmholtz resonator and parameters

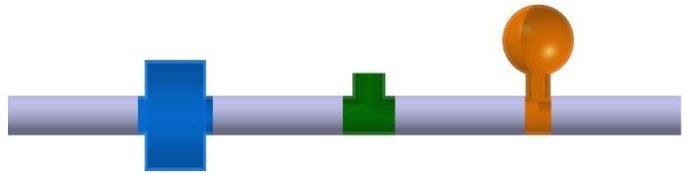
The f_{n} is finally tuned to be around 300 Hz:

$\rightarrow l_{\mathrm{H}} = 60\ \mathrm{mm},\ a_{\mathrm{H}} = 7\ \mathrm{mm},\ r_{\mathrm{H}} = 25\ \mathrm{mm}$

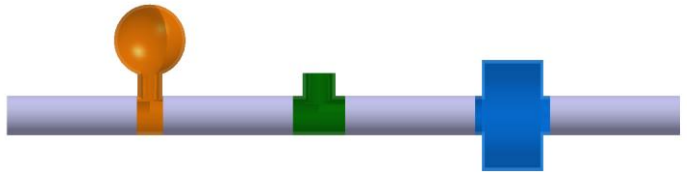
Effect of order of elements



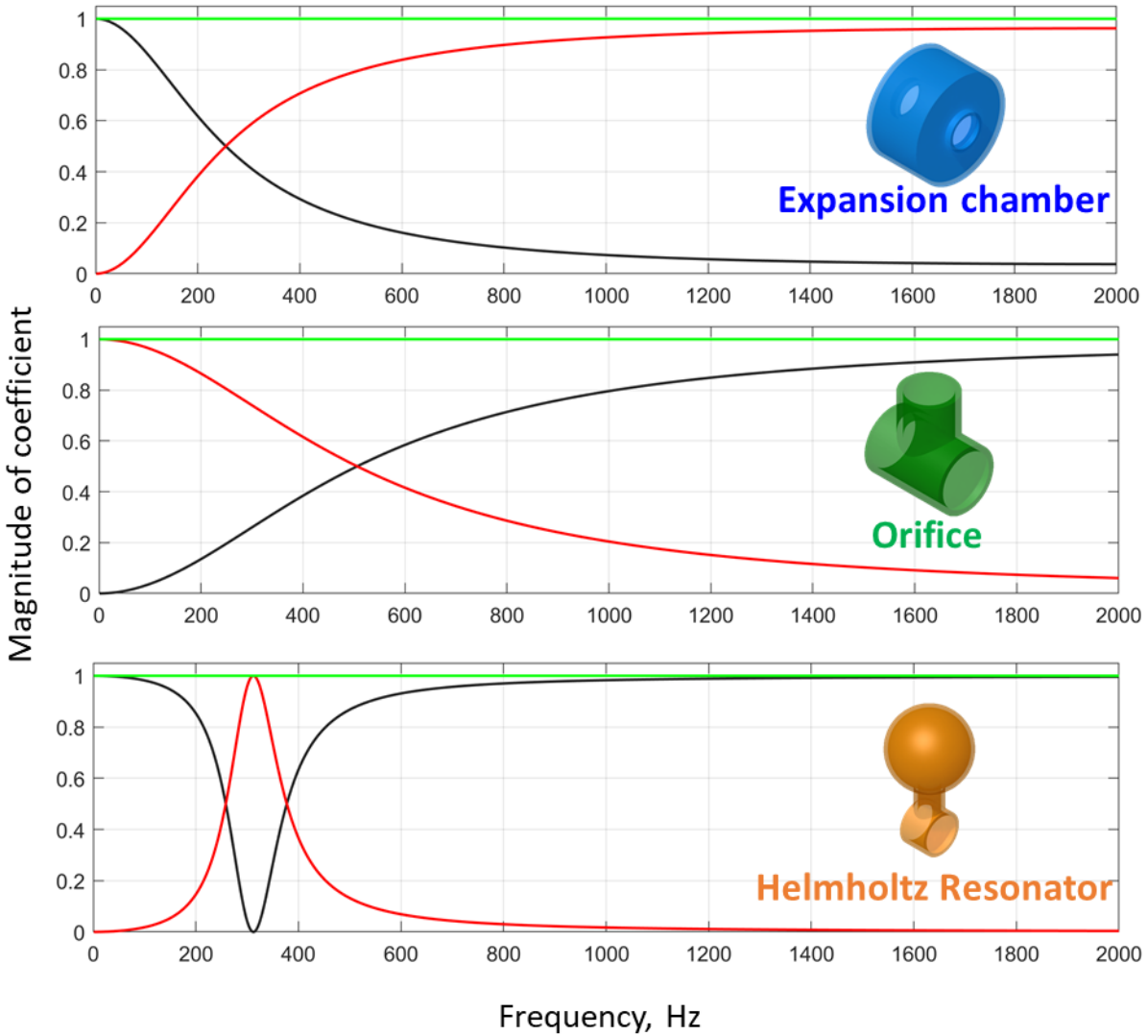
- We see 3 'sets' of results
- Why is this?
  - A filter operates the same 'forwards' and 'backwards'
  - e.g.,

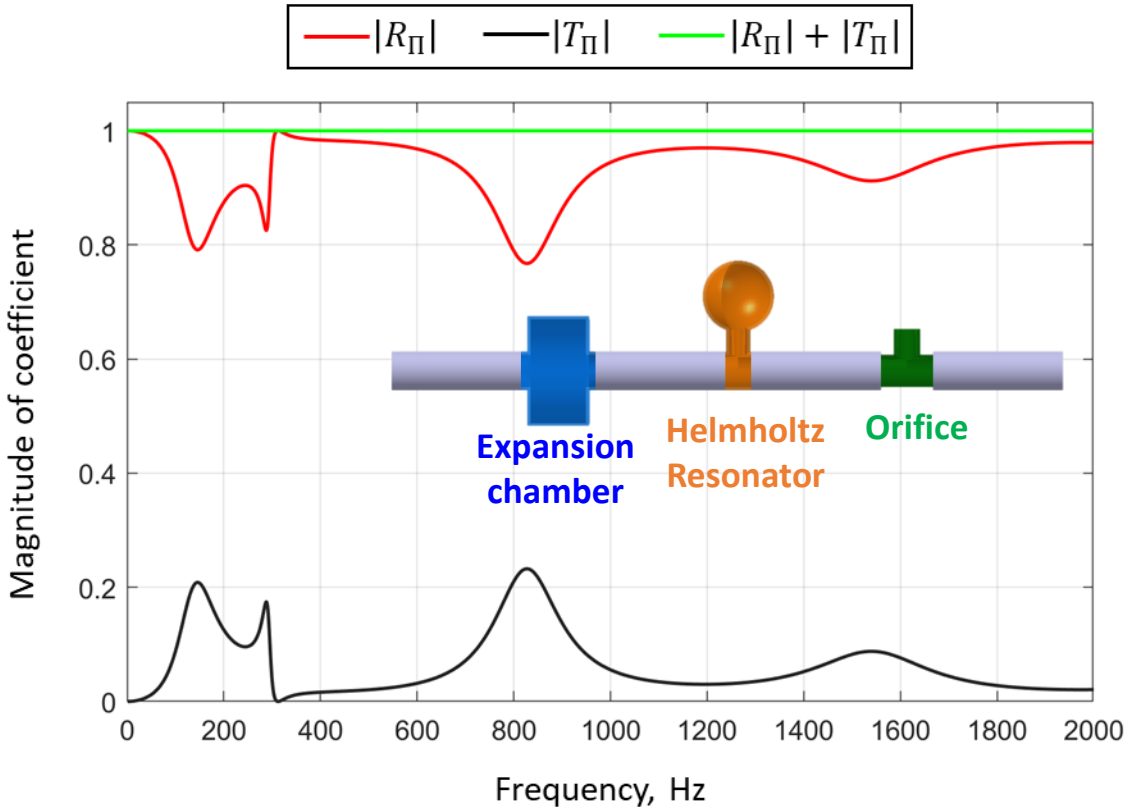


performs equivalently to

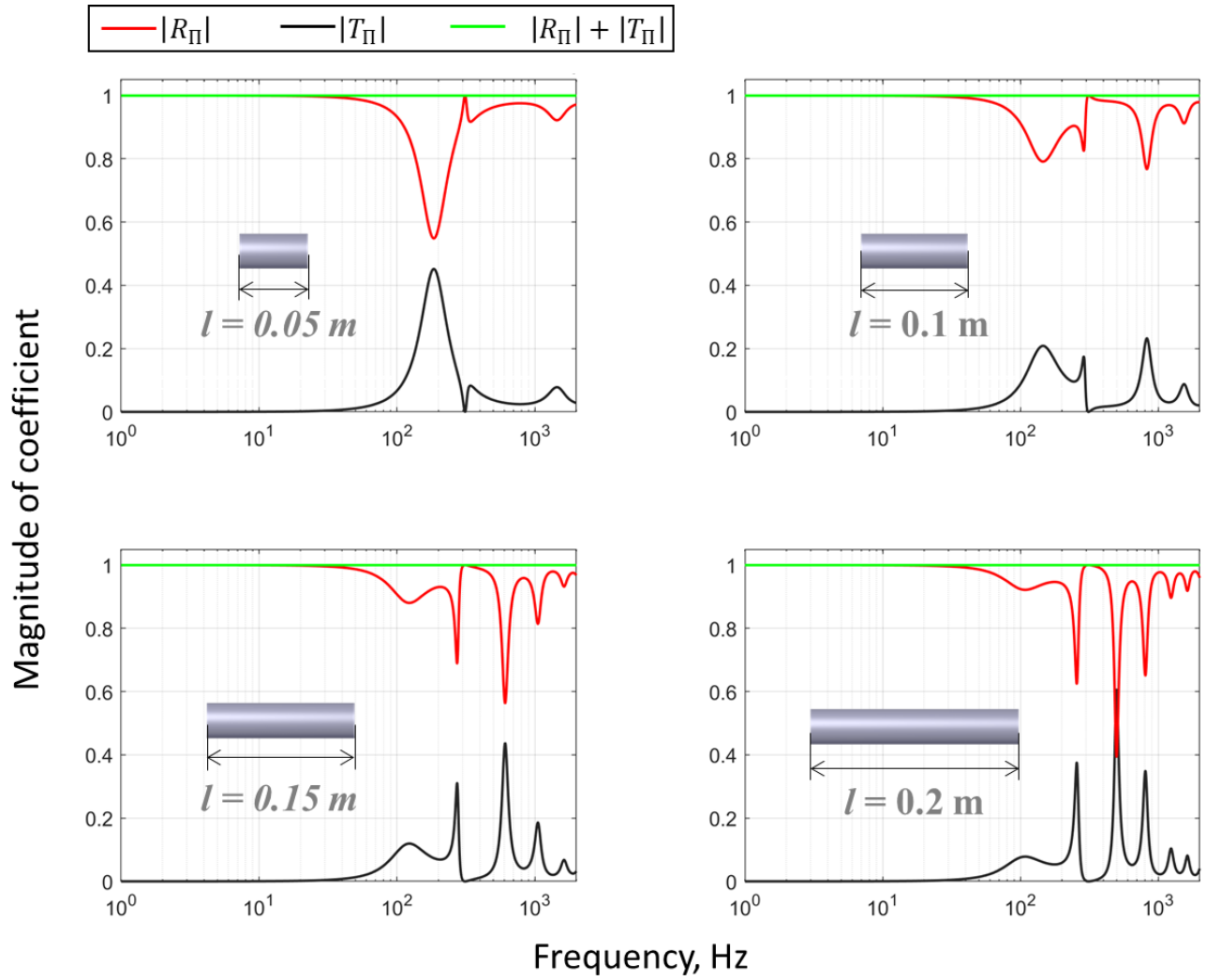


Performance of entire system (E-H-O order)





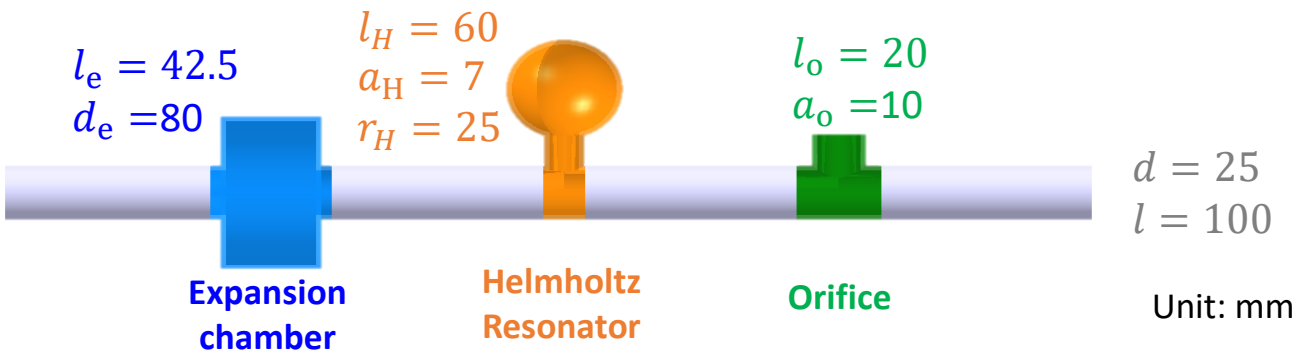
Effect of the total length of the system



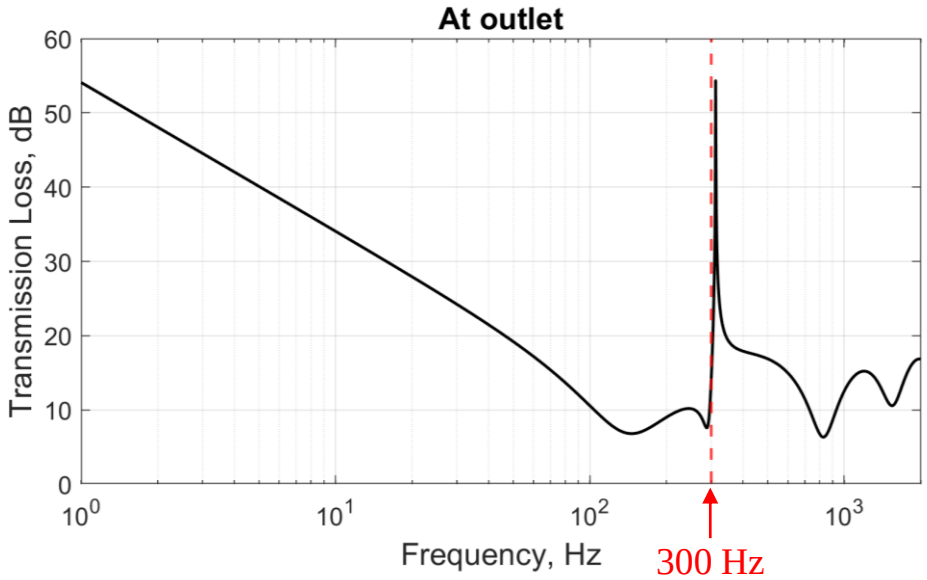
- Increasing the total length increases the number of peaks we see in the transmission coefficient

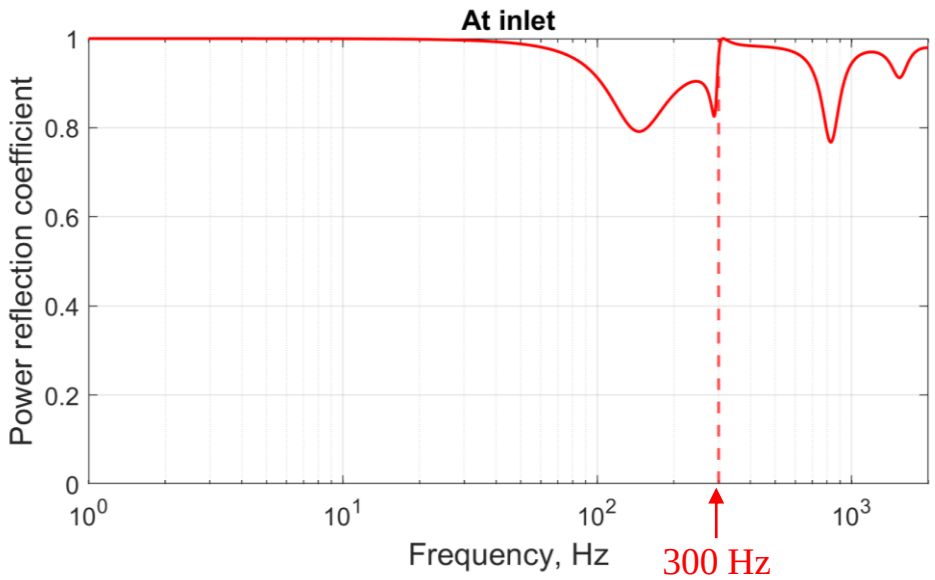
Summary & Discussion

- Objective: Design an acoustic filter to reduce noise from a temperature humidity chamber
  - Problematic 300 Hz noise produced by condenser fan
  - Also looking to attenuate the entire frequency range
- Our Filter:



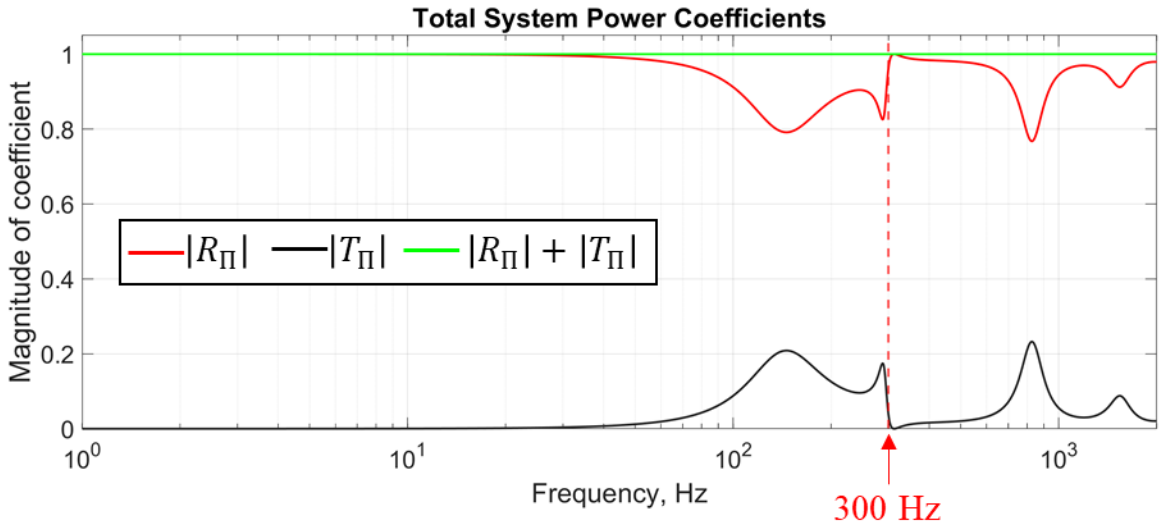
- Results:



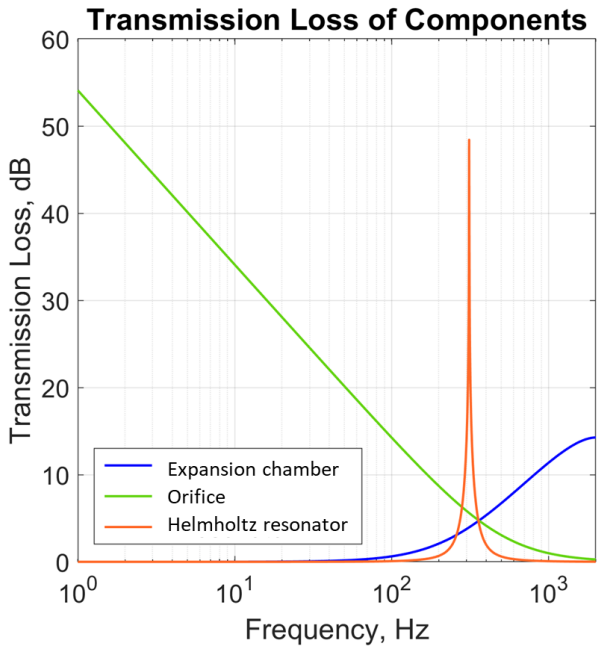


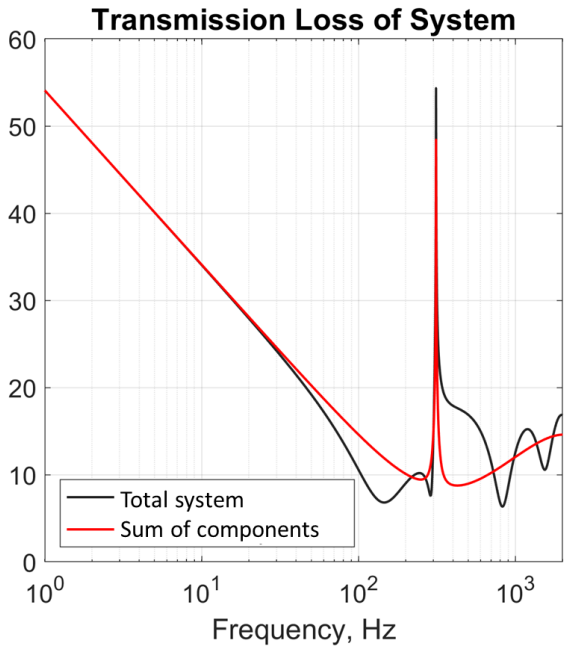
Summary & Discussion, cont.

- Verification of Results by Summation of Power Coefficients to Unity:
  - The Power Transmission and Power Reflection Coefficients do sum to 1 at all frequencies



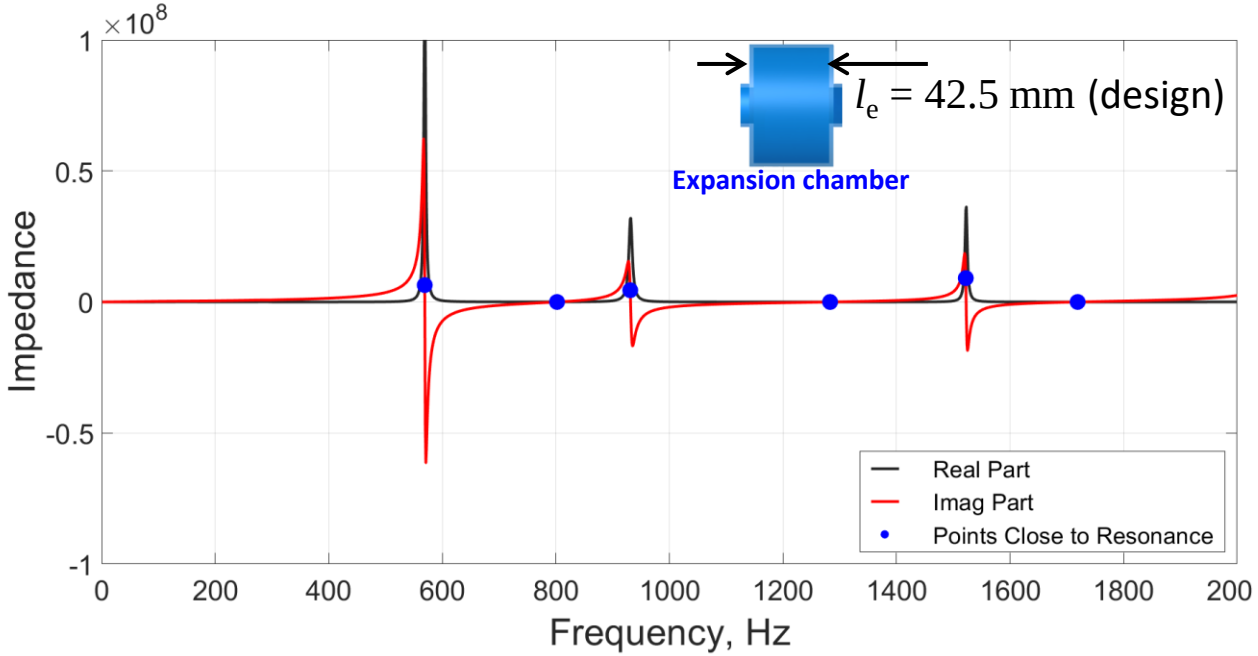
- Transmission Loss of Individual Components vs. Whole System:
  - The sum of the TL of components is NOT equal to the total TL.

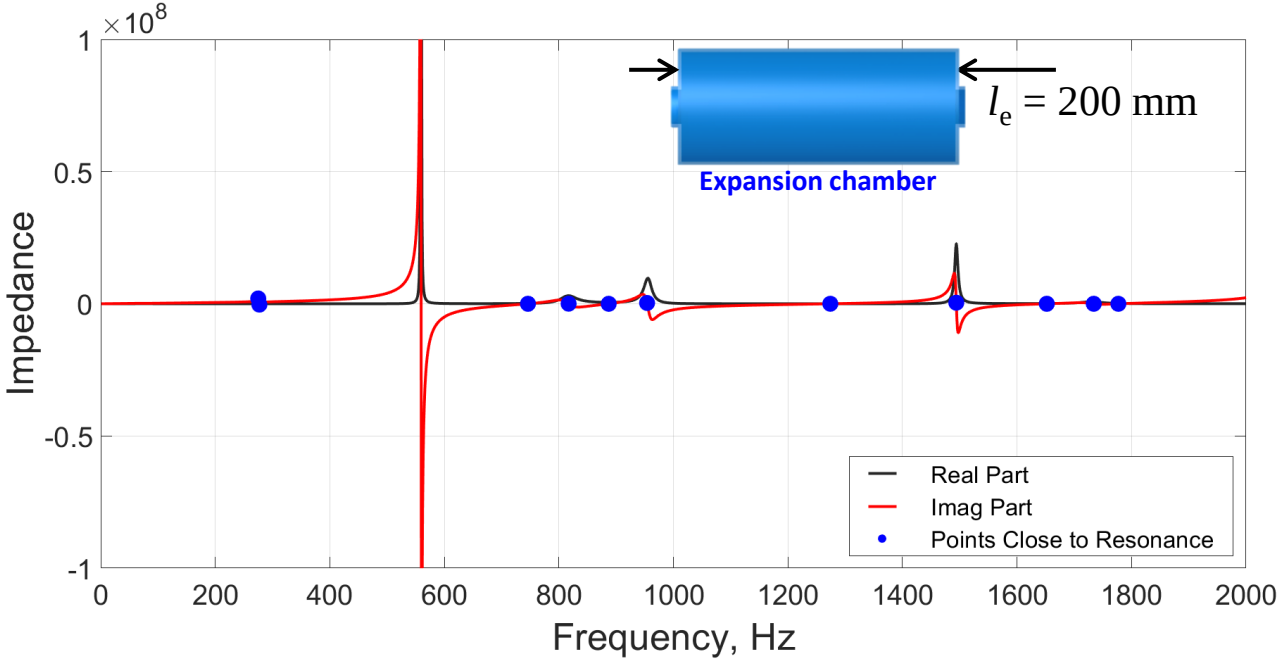




Summary & Discussion, cont.

- Input Impedance:
  - Resonance greatly influenced by length of the system
  - The resonance points can be associated with the length of the expansion chamber
  - The real impedance implies an energy loss in the system, which in this case does not represent dissipation, but rather energy leaving the system at the outlet.





Thank you for your attention!